

1. Administrative & Study Identification

Full Study Title: Efficacy Study of Nivolumab Plus Ipilimumab or Nivolumab Plus Chemotherapy Against Chemotherapy in Stomach Cancer or Stomach/Esophagus Junction Cancer **Short Title/Acronym:** CheckMate 649 (CA209-649) **Protocol Number:** CA209-649 **NCT Number:** NCT02872116 **Sponsor Name:** Bristol Myers Squibb **Investigational Product:** Nivolumab (Opdivo) plus Chemotherapy (FOLFOX or XELOX/CapeOX), and Nivolumab plus Ipilimumab (Yervoy) **Phase of Development:** Phase III (PHASE3) **Medical Monitor:** Bristol Myers Squibb Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate whether nivolumab plus chemotherapy improves overall survival (OS) or progression-free survival (PFS), as assessed by blinded independent central review (BICR), compared with chemotherapy alone in patients with previously untreated advanced or metastatic gastric cancer, gastroesophageal junction (GEJ) cancer, or esophageal adenocarcinoma whose tumors express a PD-L1 combined positive score (CPS) ≥ 5 . **Secondary Objectives:** To evaluate OS in patients with PD-L1 CPS ≥ 1 and in all randomized patients; to evaluate Objective Response Rate (ORR), Duration of Response (DOR), and to assess the efficacy of nivolumab plus ipilimumab versus chemotherapy alone.

2.2 Background & Rationale To address the critical need for improved first-line therapies and survival outcomes in patients with advanced or metastatic gastric, GEJ, and esophageal adenocarcinomas. Prior to this study, standard frontline treatment was limited to chemotherapy, which offered modest survival benefits. This study explores the addition of the PD-1 inhibitor nivolumab to standard chemotherapy (or in combination with ipilimumab) to trigger an enhanced immune response and extend survival.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Chemotherapy alone) **Blinding:** Open-label **Randomization:** Randomized (1:1 for the nivolumab+chemotherapy vs. chemotherapy arms) **Trial Status:** COMPLETED (Study has been completed)

3.2 Planned Enrollment & Duration Targeted Enrollment: 2,031 patients **Number of Sites:** Global, Multicenter **Estimated Study Start:** 12-Oct-2016 **Estimated Primary Completion:** 27-May-2020 (Actual)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or Female at least 18 years of age. 2. Must have histologically confirmed gastric cancer, gastroesophageal junction cancer, or esophageal adenocarcinoma that cannot be operated on and that is advanced or metastatic (has spread out). 3. Did not receive neoadjuvant or adjuvant treatment (chemotherapy, radiotherapy, or both) for their disease within the last 6 months. 4. Must have full activity or, if limited, must be able to walk and carry out light activities such as light house work or office work (ECOG Performance Status of 0 or 1). 5. Must agree to provide a tumor tissue sample, either from a previous surgery or biopsy within 6 months or fresh, prior to the start of treatment in this study to determine PD-L1 status.

4.2 Exclusion Criteria 1. Known HER2-positive status. 2. Presence of tumor cells in the brain or spinal cord that have not been treated. 3. Active known or suspected autoimmune disease. 4. Any serious or uncontrolled medical disorder or active infection, including known history of a positive test for human immunodeficiency virus (HIV) or known acquired immunodeficiency syndrome (AIDS). 5. Any positive test result for hepatitis B or C indicating acute or chronic infection.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Arm A (Nivolumab + Chemo):** Nivolumab 360 mg + XELOX (capecitabine and oxaliplatin) every 3 weeks, OR Nivolumab 240 mg + FOLFOX (fluorouracil [ATC Code: L01BC02; Trade names: Carac, Efudex, Atracil, Fluoroplex, Tolak], leucovorin, and oxaliplatin) every 2 weeks. * **Arm B (Nivolumab + Ipilimumab):** Nivolumab 1 mg/kg + Ipilimumab 3 mg/kg every 3 weeks for 4 cycles, followed by Nivolumab 240 mg every 2 weeks. **Route of Administration:** Intravenous (IV) for Nivolumab, Ipilimumab, FOLFOX, and Oxaliplatin; Oral for Capecitabine. **Duration of Treatment:** Until disease progression, unacceptable toxicity, withdrawal of consent, or maximum protocol-defined duration.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care medications for adverse events (e.g., antiemetics, anti-diarrheals). **Prohibited:** Prior systemic therapy for metastatic disease; concurrent investigational anticancer therapies.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Tissue for Central PD-L1 Testing
Baseline	Day 0	Randomization, First Dose Administration
Interim	Every 2 or 3 Weeks	Treatment Cycles, Safety Labs, Efficacy Assessment, irAE Monitoring
End of Study	Follow-up	Disease Progression confirmation, Survival Follow-up, IP Accountability

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Overall Survival (OS) and Progression-Free Survival (PFS) in patients whose tumors express a PD-L1 combined positive score (CPS) ≥ 5 . **Measurement Method:** PFS is defined as the time from randomization to the date of the first documented PD or death due to any cause, assessed by Blinded Independent Central Review (BICR) using RECIST v1.1 criteria; OS measured as the time from randomization to death from any cause.

7.2 Secondary & Exploratory Endpoints * Overall Survival (OS) and PFS in patients with PD-L1 CPS ≥ 1 and in all randomized patients (Intent-to-Treat population). * Objective Response Rate (ORR). * Median Duration of Response (mDOR).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection at each dosing visit. Key monitored AEs for the combination include peripheral neuropathy, nausea, fatigue, diarrhea, vomiting, decreased appetite, abdominal pain, constipation, and musculoskeletal pain. **Serious Adverse Events (SAEs):** Reported within 24 hours. Specific attention given to immune-related adverse events (irAEs) requiring corticosteroid intervention. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee assigned to evaluate ongoing safety and efficacy profiles.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for all primary and secondary efficacy evaluations (hierarchically tested in CPS ≥ 5 , then CPS ≥ 1 , then all randomized). Safety Set for all patients who received at least one dose of the assigned treatment. **Sample Size Justification:** Sized to demonstrate statistical superiority of nivolumab + chemotherapy over chemotherapy alone for OS and PFS using an $\alpha = 0.05$ boundary. **Handling of Missing Data:** Standard survival analysis

methodologies including Kaplan-Meier estimation; patients lost to follow-up or alive at the analysis cutoff were right-censored.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
Monitoring Plan: Standardized clinical monitoring including centralized PD-L1 CPS testing using the Agilent/Dako PD-L1 IHC 28-8 pharmDx test. **Audit Trail:** eCRF locking, and rigorous Blinded Independent Central Review (BICR) for all radiological tumor assessments.

11. Study Identifiers & Governance

Primary ID: CA209-649 **Posting ID:** 142177259292238040 **Registry Number:** NCT02872116 **Sponsor/Lead Organization:** Bristol Myers Squibb (Company: Bristol Myers Squibb) **Study Title:** CheckMate 649 **Official Regulatory Title:** A Randomized, Multicenter, Open-Label, Phase 3 Study of Nivolumab Plus Ipilimumab or Nivolumab in Combination With Oxaliplatin Plus Fluoropyrimidine Versus Oxaliplatin Plus Fluoropyrimidine in Subjects With Previously Untreated Advanced or Metastatic Gastric or Gastroesophageal Junction Cancer

12. Clinical Framework (Gastric/GEJ Cancer Specific)

Primary Condition: Advanced or Metastatic Gastric Cancer, Gastroesophageal Junction Cancer, and Esophageal Adenocarcinoma.
Preferred UMLS Name: Esophagogastric Cancer (Disease Sub-type: Adenocarcinoma [MeSH Code: D000230; SNOMED ID: 1187332001])
Diagnosis Threshold: Histologically/cytologically confirmed unresectable disease; centrally evaluated PD-L1 CPS
Medical Methodology: First-line immune checkpoint inhibition combined with fluoropyrimidine- and platinum-containing chemotherapy
Optimization Target: Extension of Overall Survival (OS) and Progression-Free Survival (PFS)

13. Study Procedures & Methodology

Management Pathway: Standard oncology clinical pathway incorporating immuno-oncology infusions with cytotoxic regimens.
Education Protocol (HCP): Specialized training on the early detection and management of immune-mediated adverse events (irAEs). **Education Protocol (Patient):** Treatment diaries (for patients receiving oral capecitabine), toxicity awareness, and consent. **Quality Audit Mechanism:** Centralized laboratory scoring

for PD-L1 biomarkers and independent central review for scan reads.

14. Observation & Follow-Up Schedule

Total Duration: Treatment until progression or unacceptable toxicity; long-term survival follow-up spans up to 60+ months post-randomization. **Onsite Visit Cadence:** Every 2 weeks (FOLFOX regimens) or every 3 weeks (XELOX/Ipilimumab regimens). **Remote Monitoring Frequency:** Routine adverse event and survival follow-up calls when off active treatment. **Checklist Review Interval:** Radiological tumor assessments performed every 6 weeks for the first 12 months, then every 12 weeks until disease progression.

15. Sign-off & Approval

Role	Name	Signature	Date		----	----	----	----
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					